

EXAM 7 – SPRING 2015

14. (2.5 points)

Given the following information:

Basic premium factor	0.20
Loss conversion factor	1.20
Tax multiplier	1.05
Loss capping ratio (first adjustment)	85%
Incremental loss capping ratio (second adjustment)	60%
Expected loss ratio	75%
Percent of loss emerged at first adjustment	80%

a. (1.25 points)

Calculate the premium development to loss development (PDLD) ratios for the first and second premium adjustments.

b. (0.5 point)

Briefly explain two benefits of using the retrospective rating formula method over using historical data when calculating the PDLD ratio.

c. (0.75 point)

Assuming all claims are greater than the plan minimum, explain how a push to settle small claims faster would impact PDLD ratios.

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QUESTION 14	
TOTAL POINT VALUE: 2.5	LEARNING OBJECTIVE: A14: Forecast Premium Reserves.
SAMPLE ANSWERS	
Part a: 1.25 points	
<i>Sample Answer 1</i> $\begin{aligned} \text{PDLD}(1) &= \{ \{ \text{Bp} / (\text{Sp} * \text{ELR} \% * \text{L1}) \} + \{ (\text{CL} / \text{L}) * \text{LCF} \} \} * \text{TM} \\ &= \{ \{ .2 / (.75 * .8) \} + \{ .85 * 1.2 \} \} * 1.05 \\ &= 1.421 \\ \text{PDLD}(2) &= \text{inc. cap} * \text{LCF} * \text{TM} = .6 * 1.2 * 1.05 = .756 \end{aligned}$ <i>Sample Answer 2</i> $\text{PDLD}(1) = \{ .2 + (.75 * .8 * .85 * 1.2) \} / (.8 * .75) = .756$ $\text{PDLD}(2) = .6 * 1.2 * 1.05 = .756$	
Part b: 0.5 point	
<i>Sample Answer 1</i> 1. Formula can reflect pricing parameters that are currently being sold. 2. PDLD ratios calculated by formula are more stable than those from empirical data. <i>Sample Answer 2</i> 1. Terms of policies may have changed since historical policies were written. Using the retro rating formula makes the PDLD ratios better reflect current conditions. 2. Patterns in historical data can be extremely volatile making it difficult to make development pattern selections.	
Part c: 0.75 point	
<i>Sample Answer 1</i> A push to settle small claims faster would increase the amount of claims in the early periods that fall within plan limitations since these are not subject to per occurrence limit. This would likely increase early PDLD ratios. On the other hand, settling small claims early makes later loss emergence is mostly from large claims, so the PDLD ratios for later adjustments will drop. This is because the large claim development likely occurs outside the plan parameters. <i>Sample Answer 2</i> Settling small claims faster (assuming small means below cap) will increase loss and capped loss by the same amount. So, ratio of CL/L will increase towards one. Earlier PDLD ratios will be higher. But, since later PDLD claims will now see only the larger claims (that can hit the cap) develop, those PDLD ratios will decrease. PDLD line segments get flatter faster basically, but start steeper.	
EXAMINERS' REPORT	
The candidates were expected to calculate the "Premium Development to Loss Development ratios" and the properties of these ratios under various conditions. Candidates generally scored well in part a wherein they had to calculate the ratios. However, the performance dipped when	

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they had to verbalize the relationships under different conditions in part b.
Part a
A majority of the candidates had a good understanding of this part. • The candidate was expected to know how to calculate the “Premium Development” to “Loss Development” ratio at different points in the life-cycle of a retro rated policy. • The candidate was expected to understand the data provided in the question, use the appropriate formulae, and compute the PDLD ratios at the first and second adjustment period correctly. Full credit was given if the candidates went straight to the computation without the intermediate step of writing down the formulae. • A few candidates were unable to link the appropriate meaning to the values provided in the question and used them incorrectly. • There were very few arithmetic errors.
Part b
This part was more challenging than part a. • The candidate was expected to provide a brief description of two benefits of using a formula based approach to calculate PDLD ratios as against using an empirical “loss reported data history” as the basis. • The candidate should have summarized the potential for a “responsive” method that facilitates the inclusion of the latest pricing mechanisms used in selling the policies and secondly the reduction in the volatility of the estimated ratios, when the process is driven by the “formula” method as against the “historical data” method. • Many candidates were careless about the wording of their answers and did not relate the underlying factors in a correct and coherent manner. Many candidates overlooked the fact that a time element was inevitable before the final closure of a retro treaty.